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SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE**

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Self Financed B.B.A.-B.Com. (EM & GM)-B.C.A. (Regular/AI), B.Sc. (DS), M.Sc. (IT), M.Com. (EM & GM) Degree Programme

Accredited by National Assessment and Accreditation Council with "B" Grade

Dr. Mukesh R. Goyani

M.Com., M.Ed., M.Phil., GSET, Ph.D.

Principal



At & Po. : Amroli, Surat,
Pin. 394107, Gujarat (India)

www.amrolicollege.ac.in

☎ (0261) 2494073, 2495643, 2496478

☎ 9427113947, 6354626492

✉ principal@amrolicollege.ac.in

Principal_230@vnsgu.ac.in

PROJECT REPORT

ON

Analysis of VNsgu Student Academic Performance using Python

AS

A PARTIAL REQUIREMENT FOR THE DEGREE

OF

BACHELOR OF COMPUTER APPLICATION(BCA)

FOR SUBJECT

602 DATA ANALYSIS USING PYTHON

Submitted by:

Yug Sanjaybhai Patel (6308)

Darshan Mukeshbhai Parmar (6291)

Akshat Ghanashaymbhai Sabhadiya (6354)

Guided by:

Asst. Prof. Twinkle S. Panchal

Sutex Bank College of Computer Applications and Science, Amroli



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This is to certify that Mr./Ms. SABHADIYA AKSHAT GHANASHYAMBHAI
(Exam No: 6354 & Roll No: 602) has successfully completed his/her
Minor Project Work entitled VNSGU BCA SEM-5 RESULT ANALYSIS for
the Subject: 602 - Data Analytics using Python of the 6th Semester
during the academic year 2025-26 under Sutex Bank College of
Computer Applications & Science affiliated to Veer Narmad South Gujarat
University, Surat.

Group No. : TY6-19

Internal Guide : Ms. Twinkle S. Panchal

Signature : 

Date : 30.03.2026




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External Examiner : _____



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✉ principal@amrolicollege.ac.in
Principal_230@vnsgu.ac.in

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This is to certify that Mr./Ms. PATEL YUG SANJAYBHAI (Exam No: 6308 & Roll No: 638) has successfully completed his/her Minor Project Work entitled VNSGU BCA SEM-5 RESULT ANALYSIS for the Subject: 602 - Data Analytics using Python of the 6th Semester during the academic year 2025-26 under Sutex Bank College of Computer Applications & Science affiliated to Veer Narmad South Gujarat University, Surat.

Group No. : **TY6-19**

Internal Guide : **Ms. Twinkle S. Panchal**

Signature :

Date : **30.03.2026**



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M.Com., M.Ed., M.Phil., GSET, Ph.D.
Principal

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Pin. 394107, Gujarat (India)

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9427113947, 6354626492

principal@amrolicollege.ac.in

Principal_230@vnsgu.ac.in

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This is to certify that Mr./Ms. PARMAR DARSHAN MUKESHBHAI (Exam No: 6291 & Roll No: 673) has successfully completed his/her Minor Project Work entitled VNSGU BCA SEM-5 RESULT ANALYSIS for the Subject: 602 - Data Analytics using Python of the 6th Semester during the academic year 2025-26 under Sutex Bank College of Computer Applications & Science affiliated to Veer Narmad South Gujarat University, Surat.

Group No. : TY6-19

Internal Guide : Ms. Twinkle S. Panchal

Signature :

Date : 30.03.2026



Dr. Mukesh R. Goyani
Principal

External Examiner : _____



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Acknowledgement

We would like to express our sincere gratitude to **Asst. Prof. Twinkle S. Panchal**, Sutex Bank College of Computer Applications and Science, Amroli, for her valuable guidance, continuous encouragement, and constant support throughout the development of this project titled **“Analysis of VNSGU Student Academic Performance using Python”**, submitted as a part of the subject **Data Analysis Using Python (602)**.

Her expert knowledge, constructive suggestions, and thoughtful feedback helped us understand the concepts of data analysis, exploratory data analysis, visualization techniques, and machine learning implementation more effectively. Her motivation and guidance played an important role in successfully completing this project and improving our understanding of practical data analytics using Python tools and libraries.

We are also thankful to **Sutex Bank College of Computer Applications and Science, Amroli**, for providing us with the necessary academic environment, infrastructure, and learning resources required to carry out this project work successfully. The support from the college helped us gain practical exposure to real-world data analysis techniques and strengthened our knowledge in the field of data science and analytics.

This project provided us with an opportunity to apply theoretical concepts such as data preprocessing, statistical analysis, data visualization, and machine learning clustering techniques on real academic datasets. Through this work, we were able to develop a deeper understanding of how data-driven approaches can help in identifying patterns, trends, and insights in educational data.

Finally, we would like to extend our sincere thanks to everyone who directly or indirectly supported and encouraged us during the completion of this project. Their support and motivation contributed significantly to the successful completion of this work.

Yug Sanjaybhai Patel
Darshan Mukeshbhai Parmar
Akshat Ghanashaymbhai Sabhadiya

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1. Project Definition

Analysis of VNSGU Student Academic Performance using Python

Understanding student academic performance is essential for evaluating educational quality, identifying learning gaps, and improving institutional decision-making. In this project, we analyse the VNSGU student result dataset using Python to extract meaningful academic insights through Exploratory Data Analysis (EDA).

This project focuses on analysing structured examination result data to understand performance distribution, subject-wise trends, grading patterns, and overall academic outcomes.

The primary objective of this analysis is to transform raw examination result data into actionable insights using statistical techniques and data visualization tools.

Project Objectives :

We aim to answer the following analytical questions:

1. What is the overall performance distribution of students?
2. What are the average marks obtained in each subject?
3. Which subject has the highest failure rate?
4. How are total marks distributed among students?
5. What percentage of students passed and failed?
6. Identify top-performing students based on total marks.
7. Are there any outliers in subject-wise marks?
8. What is the correlation between subjects and total marks?

2.Dataset Details

Dataset Name: VNSGU Master Student Result Dataset

Dataset Source:

<https://ums.vnsgu.net/Result/StudentResultDisplay.aspx?HtmlURL=7596,0000>

Dataset Type: Structured Excel Dataset (.xlsx)

Dataset Domain: Educational Data Analytics / Academic Performance Analysis

Dataset Description

The dataset contains academic performance records of students enrolled under VNSGU. It includes structured information related to student identification, subject-wise marks, total marks, and result status.

The dataset typically contains the following types of attributes:

- Seat Number (Unique Student Identifier)
- Student Name
- Collage Name
- Subject-wise Marks
- Total Marks
- SGPA
- Percentage
- Result Status (Pass/Fail)

3. Python Tools and Libraries Used

Python 3.10

Python is a widely used programming language in data science due to its simplicity and powerful libraries. It supports mathematical operations, data analysis, and visualization efficiently. Its easy syntax and flexibility make it suitable for handling large datasets and performing complex computations.

Pandas

Pandas is an open-source library used for data manipulation and analysis. It provides data structures like DataFrames and Series to handle tabular data efficiently. It supports operations such as data cleaning, filtering, grouping, and file handling (CSV, Excel), making it essential for data analysis.

NumPy

NumPy is a core library for numerical computations in Python. It provides fast multi-dimensional arrays and mathematical functions. It is widely used for performing calculations and serves as the foundation for libraries like Pandas and Scikit-learn.

Matplotlib

Matplotlib is a powerful library used for data visualization. It helps create graphs such as bar charts, line plots, scatter plots, and histograms. It provides customization options to present data clearly and effectively.

Seaborn

Seaborn is a high-level visualization library built on Matplotlib. It provides attractive and informative statistical plots like heatmaps, violin plots, and distribution graphs. It simplifies data visualization and is useful for exploratory data analysis.

Scikit-learn

Scikit-learn is a machine learning library used for predictive analysis. It provides tools for classification, regression, clustering, and model evaluation. It is widely used in data science for building and analysing models.

4. Understanding data

4.1: Importing necessary Python libraries :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

4.2: Creating the data frame :

```
df = pd.read_excel("D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\Data\\raw\\VNSGU_Results.xlsx")
```

4.3 display all column names :

```
print(df.columns)
```

✓ 0.0s

```
Index(['Seat Number', 'Name', 'College Name', 'Total Marks', 'SGPA', 'Result',
      'Course_Name', 'ADVANCE_WEB_TECHNOLOGY_Ext_Theory',
      'ADVANCE_WEB_TECHNOLOGY_Ext_Practical',
      'ADVANCE_WEB_TECHNOLOGY_Total_Ext', 'ADVANCE_WEB_TECHNOLOGY_Int_Theory',
      'ADVANCE_WEB_TECHNOLOGY_Int_Practical',
      'ADVANCE_WEB_TECHNOLOGY_Total_Int', 'ADVANCE_WEB_TECHNOLOGY_Total_Obt',
      'WEB_FRAMEWORK_AND_SERVICES_Ext_Theory',
      'WEB_FRAMEWORK_AND_SERVICES_Ext_Practical',
      'WEB_FRAMEWORK_AND_SERVICES_Total_Ext',
      'WEB_FRAMEWORK_AND_SERVICES_Int_Theory',
      'WEB_FRAMEWORK_AND_SERVICES_Int_Practical',
      'WEB_FRAMEWORK_AND_SERVICES_Total_Int',
      'WEB_FRAMEWORK_AND_SERVICES_Total_Obt', '.NET_TECHNOLOGY_Ext_Theory',
      '.NET_TECHNOLOGY_Ext_Practical', '.NET_TECHNOLOGY_Total_Ext',
      '.NET_TECHNOLOGY_Int_Theory', '.NET_TECHNOLOGY_Int_Practical',
      '.NET_TECHNOLOGY_Total_Int', '.NET_TECHNOLOGY_Total_Obt',
      'LINUX_OPERATING_SYSTEM(LOS)_Ext_Theory',
      'LINUX_OPERATING_SYSTEM(LOS)_Ext_Practical',
      'LINUX_OPERATING_SYSTEM(LOS)_Total_Ext',
      'LINUX_OPERATING_SYSTEM(LOS)_Int_Theory',
      'LINUX_OPERATING_SYSTEM(LOS)_Int_Practical',
      'LINUX_OPERATING_SYSTEM(LOS)_Total_Int',
      'LINUX_OPERATING_SYSTEM(LOS)_Total_Obt', 'NETWORK_TECHNOLOGY_Total_Ext',
      'NETWORK_TECHNOLOGY_Total_Int', 'NETWORK_TECHNOLOGY_Total_Obt',
      'Course_Total_Ext', 'Course_Total_Int', 'Course_Total_Obt'],
      dtype='object')
```

4.5 display top few lines of data :

```
df.head(5)
```

```

Seat Number      Name \
0      1001  ADEP SAIKRISHNA KISHORBHAI
1      1002      AHIR KRISH PRAMODKUMAR
2      1003      AHIR MEET MANSUKHBHAI
3      1004  AHIR MITKUMAR SURESHBHAI
4      1005      AHIR NILAY ARVINDBHAI

```

```

College Name Total Marks  SGPA Result \
0 V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  332 / 550  6.18  PASS
1 V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  347 / 550  6.36  PASS
2 V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  348 / 550  6.18  PASS
3 V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  347 / 550  6.36  PASS
4 V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  325 / 550  5.82  PASS

```

```

Course_Name \
0 CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
1 CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
2 CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
3 CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
4 CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)

```

```

ADVANCE_WEB_TECHNOLOGY_Ext_Theory  ADVANCE_WEB_TECHNOLOGY_Ext_Practical \
0                                11.0                                22.0
1                                 9.0                                19.0
2                                16.0                                19.0
...
3                                25.0                                14          39.0
4                                20.0                                11          31.0

```

[5 rows x 41 columns]

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#)...

4.6 display shape of data :

```
df.shape
```

```
(8987, 49)
```

4.7 Getting summary of the data frame using df.info() :

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8987 entries, 0 to 8986
Data columns (total 41 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Seat Number                               8987 non-null   int64
1   Name                                       8987 non-null   object
2   College Name                             8987 non-null   object
3   Total Marks                             8987 non-null   object
4   SGPA                                       8987 non-null   object
5   Result                                    8987 non-null   object
6   Course_Name                             8987 non-null   object
7   ADVANCE_WEB_TECHNOLOGY_Ext_Theory        8950 non-null   float64
8   ADVANCE_WEB_TECHNOLOGY_Ext_Practical     8805 non-null   float64
9   ADVANCE_WEB_TECHNOLOGY_Total_Ext        8793 non-null   float64
10  ADVANCE_WEB_TECHNOLOGY_Int_Theory         8987 non-null   int64
11  ADVANCE_WEB_TECHNOLOGY_Int_Practical     8942 non-null   float64
12  ADVANCE_WEB_TECHNOLOGY_Total_Int         8942 non-null   float64
13  ADVANCE_WEB_TECHNOLOGY_Total_Obt         8987 non-null   int64
14  WEB_FRAMEWORK_AND_SERVICES_Ext_Theory    8948 non-null   float64
15  WEB_FRAMEWORK_AND_SERVICES_Ext_Practical  8830 non-null   float64
16  WEB_FRAMEWORK_AND_SERVICES_Total_Ext     8819 non-null   float64
17  WEB_FRAMEWORK_AND_SERVICES_Int_Theory    8986 non-null   float64
18  WEB_FRAMEWORK_AND_SERVICES_Int_Practical  8940 non-null   float64
19  WEB_FRAMEWORK_AND_SERVICES_Total_Int     8940 non-null   float64
...
40  Course_Total_Obt                         8986 non-null   float64
dtypes: float64(31), int64(4), object(6)
```

5. Understanding Spread of data

With our dataset we have used histogram with “kde=True” to understand the distribution and spread of data in column “rate” (user ratings).

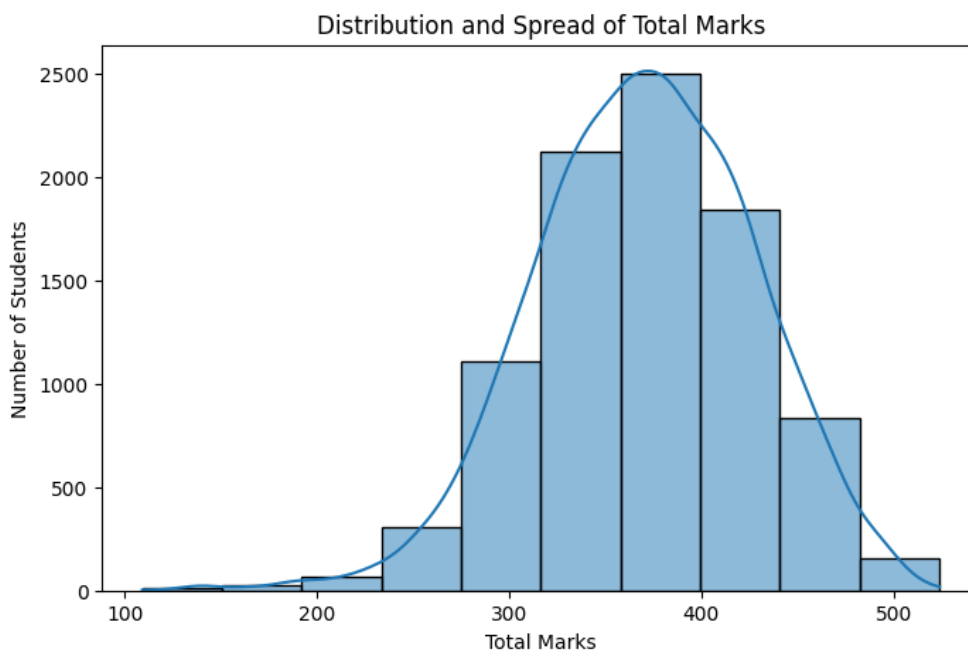
5.1 Distribution of Total Marks (Histogram)

To understand how students performed overall, we visualize the distribution of total marks using a histogram with kernel density estimation (KDE).

A histogram shows:

- How marks are distributed
- Which score range has maximum students
- Whether distribution is normal, skewed, or uneven

```
plt.figure(figsize=(8,5))
sns.histplot(df['Total'], bins=10, kde=True)
plt.title("Distribution and Spread of Total Marks")
plt.xlabel("Total Marks")
plt.ylabel("Number of Students")
plt.show()
```



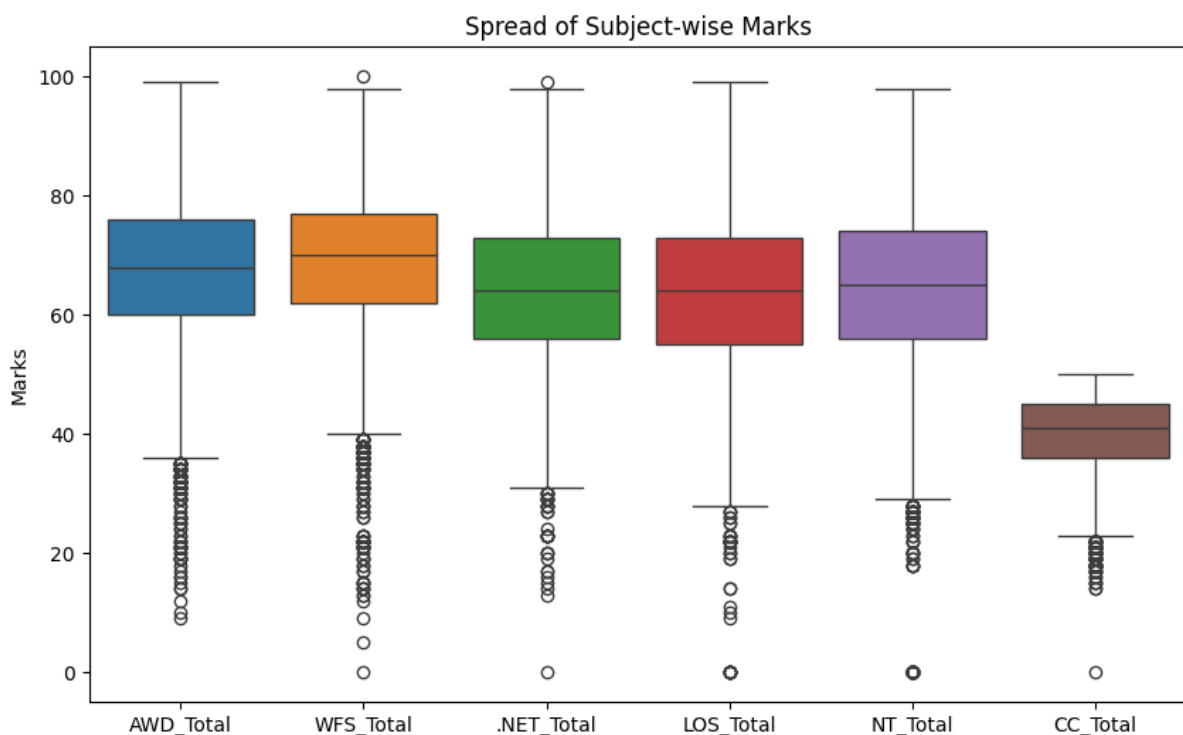
5.2 Spread of Subject-wise Marks (Boxplot Analysis)

To understand variability and detect outliers in individual subjects, we use boxplots.

A boxplot helps in:

- Identifying median marks
- Observing spread (IQR)
- Detecting outliers
- Comparing subject difficulty levels

```
plt.figure(figsize=(10,6))
sns.boxplot(data=df[['AWD_Total','WFS_Total','NET_Total','LOS_Total','NT_Total','CC_Total']])
plt.title("Spread of Subject-wise Marks")
plt.ylabel("Marks")
plt.show()
```



6. Automating EDA using python

Automating Exploratory Data Analysis (EDA) in Python is achieved using specialized libraries that generate comprehensive, interactive reports with just a few lines of code.

In Python's pandas library, `df.info()` and `df.describe()` are fundamental methods for performing initial exploratory data analysis (EDA).

`df.info()`: Provides a concise structural summary of the DataFrame. It provides following information about dataset:

- Data Types (Dtypes),
- Non-Null Counts,
- Column Names,
- Memory Usage,
- Total Entries/Rows

`df.describe()`: Offers a statistical summary of the numerical data.

- Count: The number of non-null observations in each column.
- Mean: The average value.
- Std: The standard deviation (spread of data).
- Min: The minimum value.
- 25%, 50%, 75%: The first quartile, median (50th percentile), and third quartile.
- Max: The maximum value.

These functions help data analysts quickly understand the dataset's composition and spot potential issues like missing values or outliers without manually sifting through rows.

6.1 Getting summary of the dataframe using df.info()

```
print("----- Dataset Structural Summary -----")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8987 entries, 0 to 8986
Data columns (total 49 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Seat Number                               8987 non-null   int64
1   Name                                       8987 non-null   object
2   College Name                             8987 non-null   object
3   Total Marks                              8987 non-null   object
4   SGPA                                       8987 non-null   object
5   Result                                    8987 non-null   object
6   Course_Name                              8987 non-null   object
7   ADVANCE_WEB_TECHNOLOGY_Ext_Theory         8950 non-null   float64
8   ADVANCE_WEB_TECHNOLOGY_Ext_Practical       8805 non-null   float64
9   ADVANCE_WEB_TECHNOLOGY_Total_Ext          8793 non-null   float64
10  ADVANCE_WEB_TECHNOLOGY_Int_Theory          8987 non-null   int64
11  ADVANCE_WEB_TECHNOLOGY_Int_Practical       8942 non-null   float64
12  ADVANCE_WEB_TECHNOLOGY_Total_Int           8942 non-null   float64
13  ADVANCE_WEB_TECHNOLOGY_Total_Obt           8987 non-null   int64
14  WEB_FRAMEWORK_AND_SERVICES_Ext_Theory      8948 non-null   float64
15  WEB_FRAMEWORK_AND_SERVICES_Ext_Practical    8830 non-null   float64
16  WEB_FRAMEWORK_AND_SERVICES_Total_Ext       8819 non-null   float64
17  WEB_FRAMEWORK_AND_SERVICES_Int_Theory      8986 non-null   float64
18  WEB_FRAMEWORK_AND_SERVICES_Int_Practical    8940 non-null   float64
19  WEB_FRAMEWORK_AND_SERVICES_Total_Int       8940 non-null   float64
...
47  Course_Total_Int                           8987 non-null   int64
48  Course_Total_Obt                           8986 non-null   float64
```

6.2 Getting description of the dataframe using df.describe()

```
print("----- Statistical Summary -----")
print(df.describe())
```

```

count      Seat Number  Total Marks      SGPA  AWD_Ext_Th  AWD_Ext_Pr  \
count      8987.000000  8987.000000  8987.000000  8987.000000  8987.000000
mean       5520.181262   370.020363    6.468580   12.560588   17.758874
std        2609.081077    57.192697    1.689834    4.646001    4.240496
min        1001.000000   109.000000    0.000000    0.000000    0.000000
25%        3264.500000   332.000000    6.000000    9.000000   16.000000
50%        5513.000000   371.000000    6.730000   13.000000   18.000000
75%        7771.500000   410.000000    7.450000   16.000000   20.000000
max        10042.000000  524.000000    9.820000   25.000000   25.000000

count      AWD_Ext_Total  AWD_Int_Th  AWD_Int_Pr  AWD_Int_Total  AWD_Total  \
count      8987.000000  8987.000000  8987.000000  8987.000000  8987.000000
mean         30.148103   17.628463   19.849227   37.415266   67.823968
std          7.848625    3.600709    3.735655    6.508946   11.734305
min          0.000000    5.000000    0.000000    0.000000    9.000000
25%         26.000000   15.000000   18.000000   34.000000   60.000000
50%         31.000000   18.000000   20.000000   38.000000   68.000000
75%         35.000000   20.000000   23.000000   42.000000   76.000000
max         49.000000   25.000000   25.000000   50.000000   99.000000

count      ...  LOS_Int_Pr  LOS_Int_Total  LOS_Total  NT_Ext_Total  \
count      ...  8987.000000  8987.000000  8987.000000  8987.000000
mean      ...  17.958496    35.452988    63.905530    28.364527
std      ...  4.100120     6.918373    12.854728     9.659826
min      ...  0.000000     0.000000     0.000000     0.000000
...
75%      74.545455
max      95.272727

```

7. Handling missing data and outliers data

Before performing deeper analysis, it is essential to clean and prepare the dataset. Raw academic data may contain:

- Missing values
- Incorrect data types
- Duplicate records
- Extreme values (outliers)

Data cleaning ensures accurate and reliable analysis results.

7.1 Handling Missing Values :

Missing values can distort statistical calculations and visualizations. Therefore, we first identify and handle them properly.

```
print("----- Checking Missing Values -----")
print(df.isnull().sum())
```

```
Seat Number      0
Name             0
College Name     0
Total Marks     0
SGPA            0
Result          0
Course_Name      0
ADVANCE_WEB_TECHNOLOGY_Ext_Theory    37
ADVANCE_WEB_TECHNOLOGY_Ext_Practical 182
ADVANCE_WEB_TECHNOLOGY_Total_Ext     194
ADVANCE_WEB_TECHNOLOGY_Int_Theory    0
ADVANCE_WEB_TECHNOLOGY_Int_Practical 45
ADVANCE_WEB_TECHNOLOGY_Total_Int     45
ADVANCE WEB TECHNOLOGY Total Obt     0
```

```

WEB_FRAMEWORK_AND_SERVICES_Ext_Theory      39
WEB_FRAMEWORK_AND_SERVICES_Ext_Practical    157
WEB_FRAMEWORK_AND_SERVICES_Total_Ext        168
WEB_FRAMEWORK_AND_SERVICES_Int_Theory        1
WEB_FRAMEWORK_AND_SERVICES_Int_Practical     47
WEB_FRAMEWORK_AND_SERVICES_Total_Int         47
WEB_FRAMEWORK_AND_SERVICES_Total_Obt        1
.NET_TECHNOLOGY_Ext_Theory                   39
.NET_TECHNOLOGY_Ext_Practical                103
.NET_TECHNOLOGY_Total_Ext                    113
.NET_TECHNOLOGY_Int_Theory                   1
...
NETWORK_TECHNOLOGY_Total_Obt                 49
Course_Total_Ext                             31
Course_Total_Int                             0
Course_Total_Obt                             1
dtype: int64

```

7.2: Fill Missing Values :

7.2.1 Handling Missing Marks :

During data inspection, certain subject columns contained missing values. Upon verification with examination records, these missing values represented students who scored zero marks in specific components (e.g., absent).

```

# Convert selected columns to numeric first
df[columns_to_fill] = df[columns_to_fill].apply(pd.to_numeric, errors='coerce')

# Then fill NaN with 0
df[columns_to_fill] = df[columns_to_fill].fillna(0)

```

7.2.2 Handling SGPA Column :

The SGPA column contained the value "--" for students who failed the examination. Since this does not represent zero SGPA but rather an undefined grade point, these values were converted to NaN instead of zero to maintain statistical integrity.

```
df['SGPA'] = df['SGPA'].replace('--', np.nan)
df['SGPA'] = pd.to_numeric(df['SGPA'], errors='coerce')
```

7.2.3 Cleaning Total Marks Column :

The "Total Marks" column contained values in the format "Obtained Marks / Maximum Marks" (e.g., 345/550). Since only the obtained marks were required for analysis, the denominator part was removed.

```
df['Total Marks'] = df['Total Marks'].astype(str).str.split('/').str[0].str.strip()
df['Total Marks'] = pd.to_numeric(df['Total Marks'], errors='coerce')
```

7.2.4 Converting Marks Columns to Numeric :

The subject mark columns contained some non-numeric values and blank entries. Since these represented either absent students or zero marks, they were converted into numeric format and missing values were replaced with 0.

```
df[marks_columns] = (
    df[marks_columns]
    .apply(pd.to_numeric, errors='coerce')
    .fillna(0)
    .astype(int)
)
```


8. Univariate Analysis

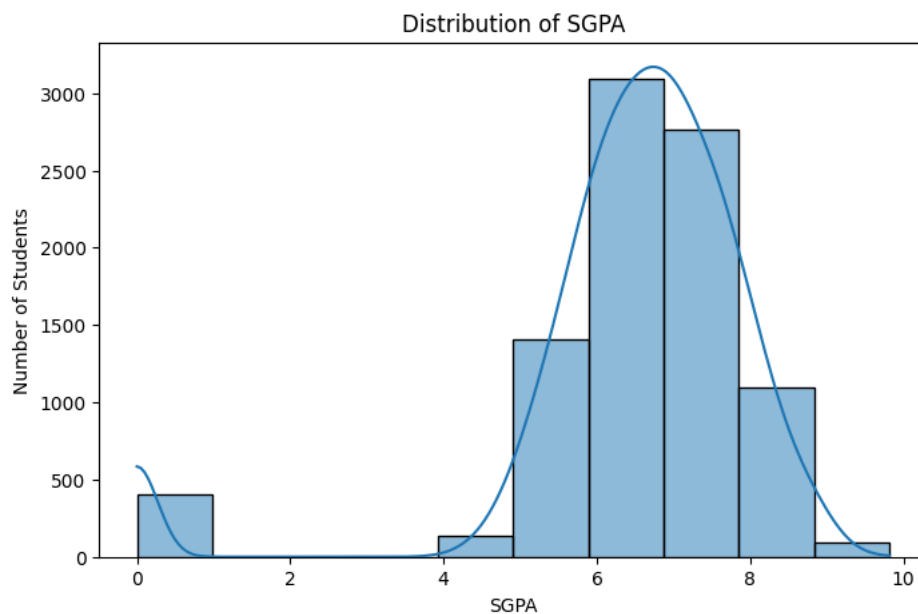
Univariate Analysis involves analysing a single variable at a time to understand its distribution, central tendency, and frequency patterns.

In this chapter, we analyse:

- SGPA Distribution
- Pass vs Fail Ratio
- Subject-wise Failure Count
- Percentage Distribution

8.1 SGPA Distribution :

```
plt.figure(figsize=(8,5))
sns.histplot(df['SGPA'], bins=10, kde=True)
plt.title("Distribution of SGPA")
plt.xlabel("SGPA")
plt.ylabel("Number of Students")
plt.show()
```

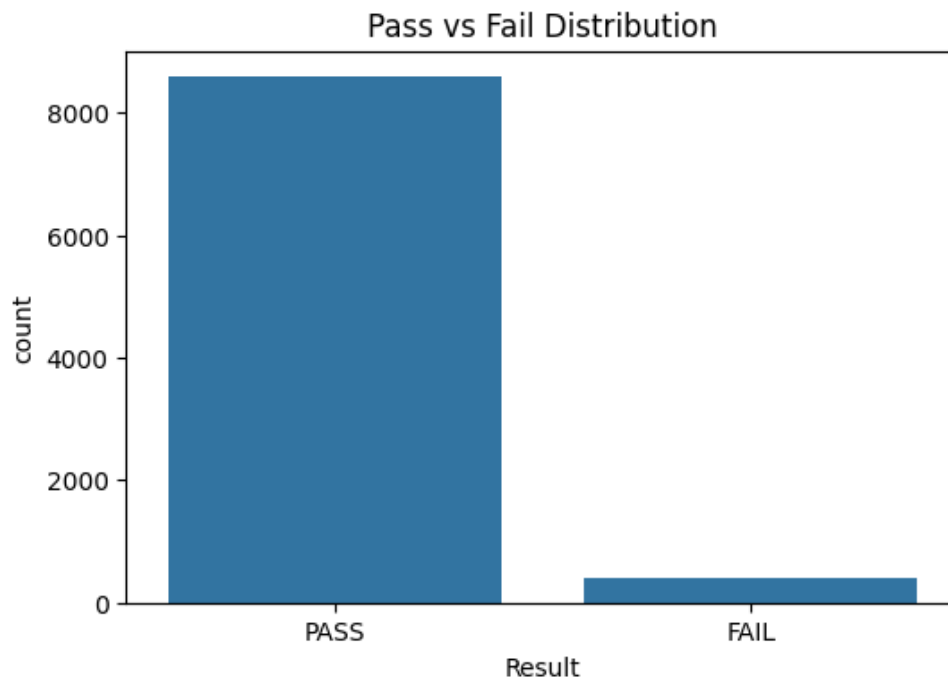


Interpretation:

- Majority clustered around 6–8 average performance.
- High peak near 8–9 strong academic performance.
- Values near 0 indicate failed students (if included).

8.2 Pass vs Fail Distribution :

```
plt.figure(figsize=(6,4))
sns.countplot(x='Result', data=df)
plt.title("Pass vs Fail Distribution")
plt.show()
```



Interpretation:

- Higher PASS count → strong semester performance.
- Low FAIL/ATKT count → difficult subjects.

8.3 Failure Count Per Subject :

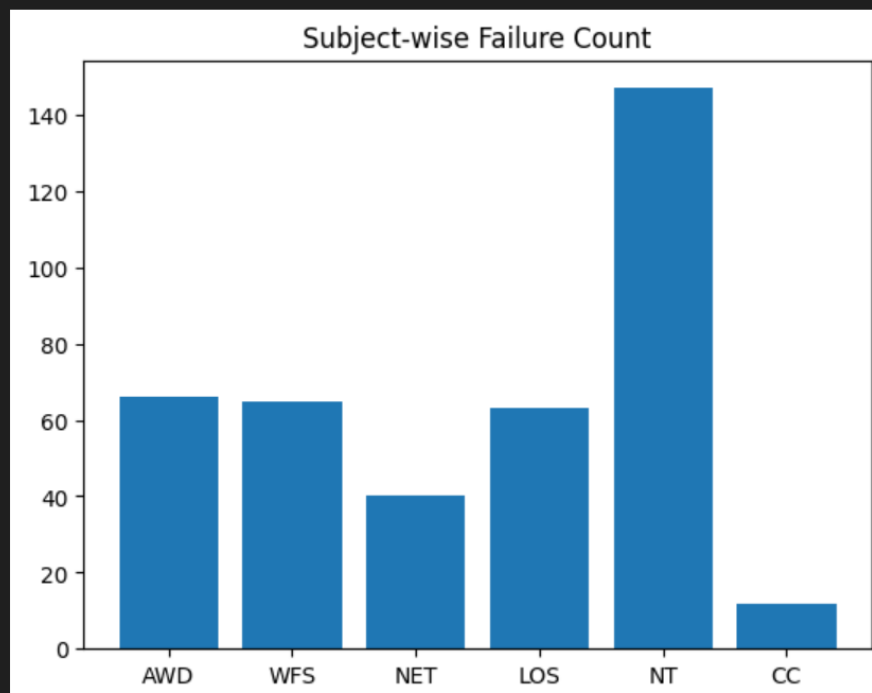
```
import matplotlib.pyplot as plt

# Subject-wise Failure Count based on Passing Criteria
failure_counts = {
    'AWD': (df['AWD_Total'] < 33).sum(),
    'WFS': (df['WFS_Total'] < 33).sum(),
    'NET': (df['.NET_Total'] < 33).sum(),
    'LOS': (df['LOS_Total'] < 33).sum(),
    'NT': (df['NT_Total'] < 33).sum(),
    'CC': (df['CC_Total'] < 18).sum(),
}

# Print Failure Counts
print("Subject-wise Failure Count:")
for subject, count in failure_counts.items():
    print(f"{subject}: {count}")

# Plot Bar Graph
plt.figure(figsize=(8,5))
plt.bar(failure_counts.keys(), failure_counts.values())
plt.title("Subject-wise Failure Count")
plt.xlabel("Subjects")
plt.ylabel("Number of Failed Students")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
{'AWD': 66, 'WFS': 65, 'NET': 40, 'LOS': 63, 'NT': 147, 'CC': 12}
```

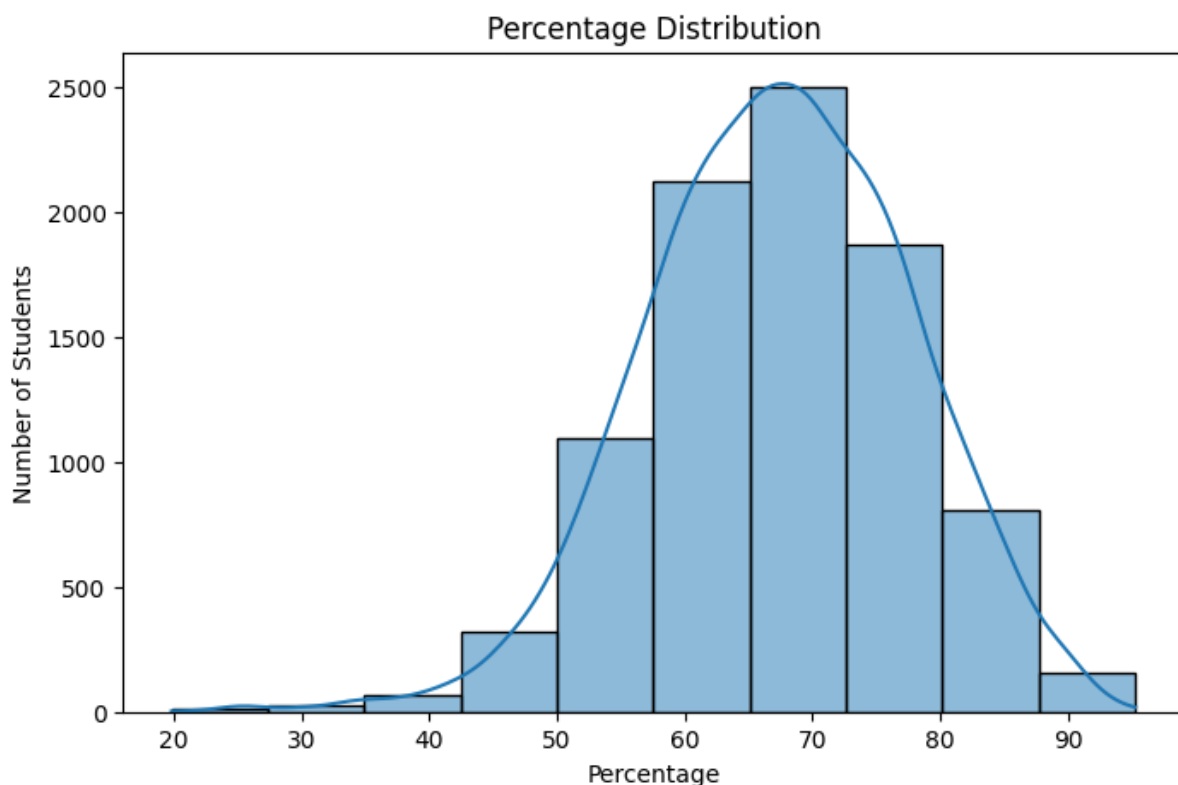


Interpretation:

- Network Technology (NT) has the highest failures (147), indicating it is the most difficult subject.
- AWD, WFS, and LOS show moderate failure counts, suggesting average difficulty.
- Cloud Computing (CC) has the lowest failures (12), showing strongest student performance.

8.4 Percentage Distribution :

```
plt.figure(figsize=(8,5))
sns.histplot(df['Percentage'], bins=10, kde=True)
plt.title("Percentage Distribution")
plt.xlabel("Percentage")
plt.ylabel("Number of Students")
plt.show()
```



Interpretation :

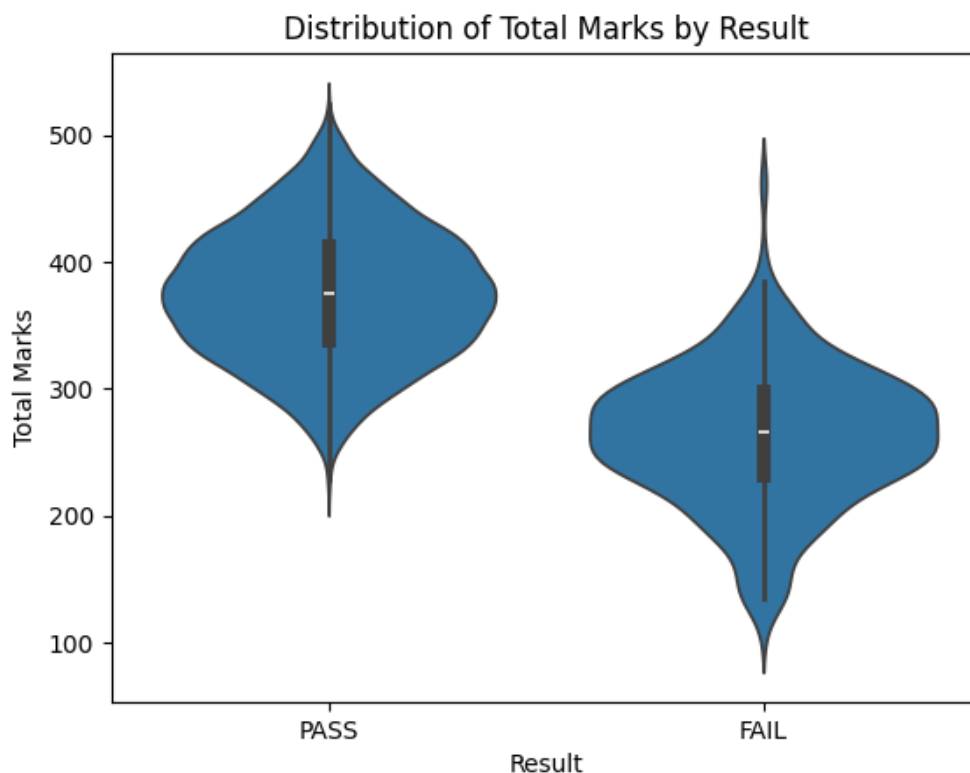
- Most students scored between 55% and 75%, indicating average to good overall performance.
- The distribution appears approximately bell-shaped (normal), showing balanced academic results.
- Very few students scored below 40% or above 90%, indicating limited extreme performance cases.

9. Bivariate Analysis

Bivariate Analysis involves examining the relationship between two variables to understand patterns, dependencies, and performance trends within the dataset. In this chapter, we analyse relationships between academic metrics such as total marks, percentage, result status, subject performance, and institutional comparison.

9.1 Distribution of Total Marks by Result :

```
sns.violinplot(x='Result', y='Total Marks', data=df)
plt.title("Distribution of Total Marks by Result")
plt.show()
```



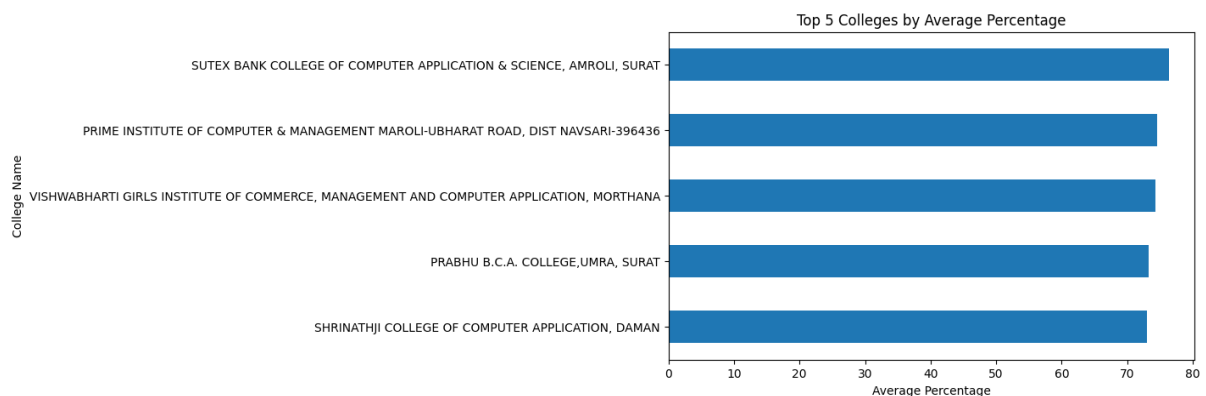
Interpretation:

The violin plot shows a clear distinction between PASS and NOT PASS students. Students who passed are concentrated in the higher total marks range, while non-passing students are clustered in the lower score region. The separation confirms that total marks strongly influence final result status.

9.2 Top 5 Colleges by Average Percentage :

```
college_avg = df.groupby('College Name')['Percentage'].mean().sort_values(ascending=False)
top5 = college_avg.head(5)

plt.figure(figsize=(8,5))
top5.sort_values().plot(kind='barh')
plt.title("Top 5 Colleges by Average Percentage")
plt.xlabel("Average Percentage")
plt.tight_layout()
plt.show()
```



Interpretation:

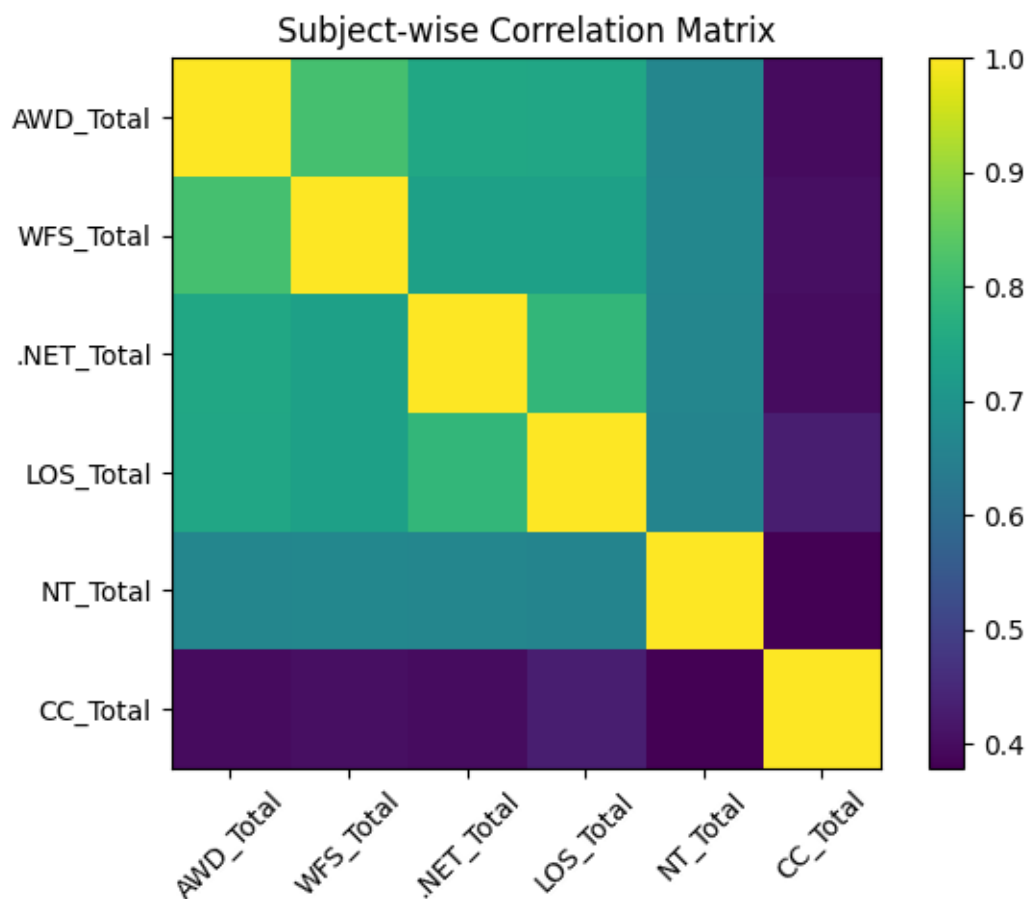
The comparison reveals measurable differences in academic performance across institutions. The top-performing colleges demonstrate consistently higher average percentages, indicating stronger academic outcomes. This analysis highlights institutional variation under the same university framework.

9.3 Subject-wise Correlation Analysis :

```
subject_cols = ['AWD_Total', 'WFS_Total', '.NET_Total', 'LOS_Total', 'NT_Total', 'CC_Total']

corr_matrix = df[subject_cols].corr()

plt.figure()
plt.imshow(corr_matrix)
plt.xticks(range(len(subject_cols)), subject_cols, rotation=45)
plt.yticks(range(len(subject_cols)), subject_cols)
plt.title("Subject-wise Correlation Matrix")
plt.colorbar()
plt.show()
```



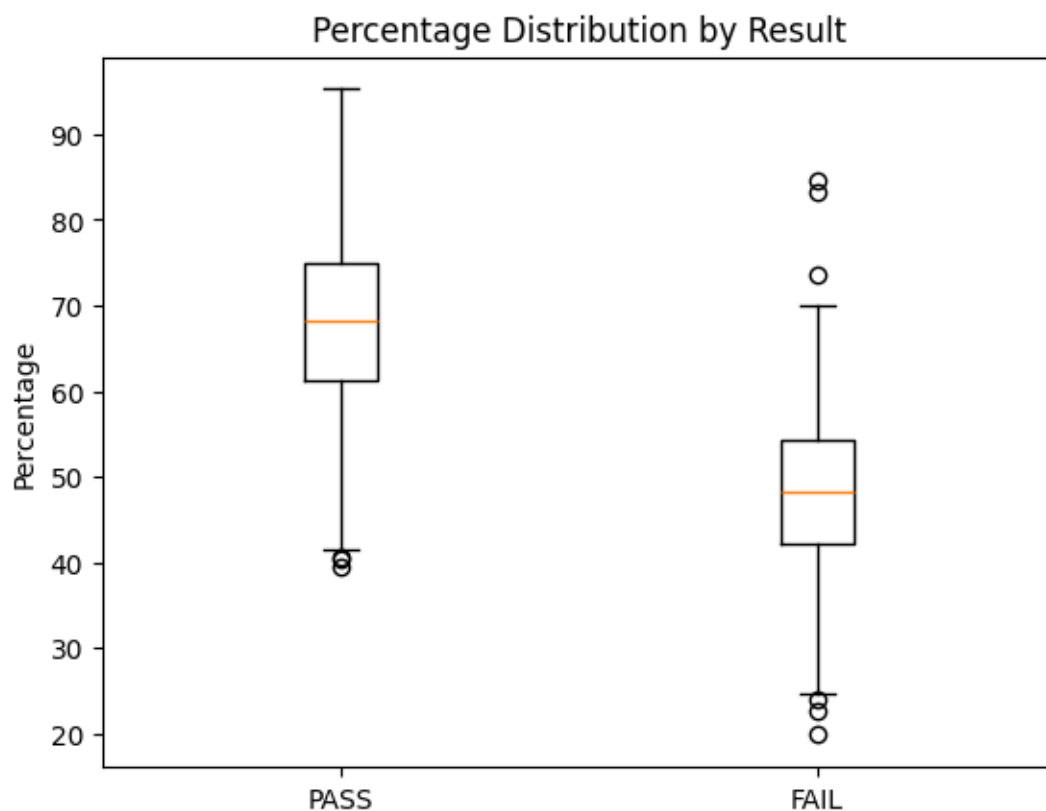
Interpretation:

The correlation heatmap indicates strong positive relationships among several theory-based subjects. High correlation values suggest that students who perform well in one subject tend to perform well in related subjects. Lower correlation values indicate relatively independent performance patterns.

9.4 Percentage Distribution by Result :

```
pass_percent = df[df['Result'] == 'PASS']['Percentage']
fail_percent = df[df['Result'] != 'PASS']['Percentage']

plt.boxplot([pass_percent, fail_percent], labels=['PASS', 'NOT PASS'])
plt.title("Percentage Distribution by Result")
plt.ylabel("Percentage")
plt.show()
```



Interpretation:

The median percentage of PASS students is significantly higher than that of NOT PASS students. The boxplot clearly shows that percentage is a strong determinant of final academic outcome. The variability among failing students is greater, indicating inconsistent performance patterns.

9.5 Subject-wise Passing Percentage :

```
# Total number of students
total_students = len(df)

# Calculate passing percentage per subject
passing_percentage = {
    'AWD': (df['AWD_Total'] >= 33).sum() / total_students * 100,
    'WFS': (df['WFS_Total'] >= 33).sum() / total_students * 100,
    'NET': (df['.NET_Total'] >= 33).sum() / total_students * 100,
    'LOS': (df['LOS_Total'] >= 33).sum() / total_students * 100,
    'NT': (df['NT_Total'] >= 33).sum() / total_students * 100,
    'CC': (df['CC_Total'] >= 18).sum() / total_students * 100
}

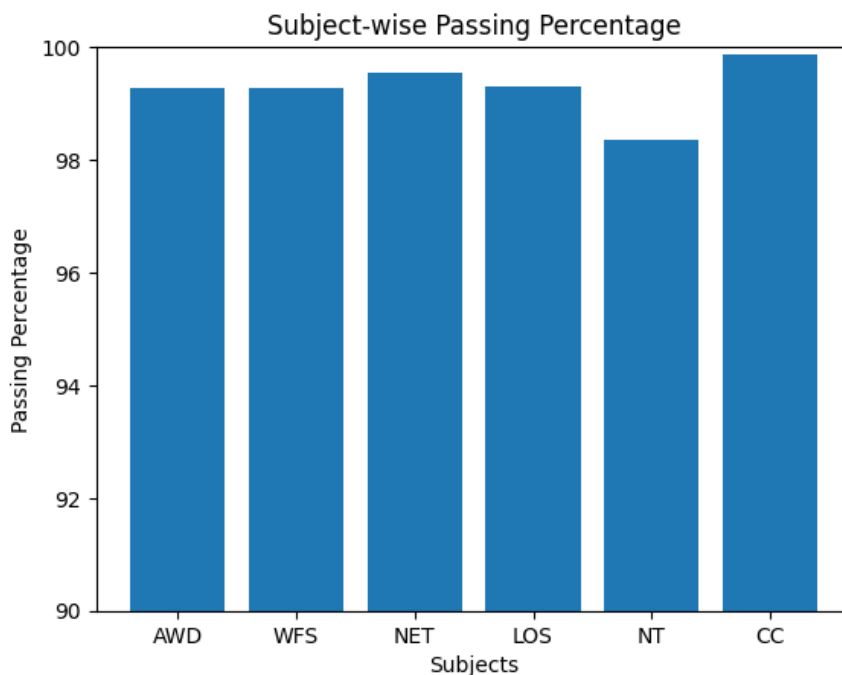
plt.figure()

plt.bar(passing_percentage.keys(), passing_percentage.values())
```

```
plt.title("Subject-wise Passing Percentage")
plt.xlabel("Subjects")
plt.ylabel("Passing Percentage")

# Zoom into the important range
plt.ylim(90, 100)
plt.yticks(range(90, 101, 2))

plt.show()
```



Interpretation:

Most subjects demonstrate high passing percentages, indicating strong overall academic performance. However, comparatively lower passing rates in certain subjects highlight potential areas of academic difficulty. This analysis provides valuable insight for curriculum review and academic improvement strategies.

10. Multivariate Analysis

Multivariate analysis examines the relationship among three or more variables simultaneously to identify complex patterns within the dataset. In the context of student result analysis, multiple academic indicators such as subject marks, SGPA, percentage, and result status are analysed together to understand how different factors collectively influence student performance. This approach provides deeper insights compared to univariate or bivariate analysis by considering multiple dimensions of academic data at the same time.

10.1 Subject-wise Internal vs External Average Marks :

```
subjects = ['AWD', 'WFS', 'NET', 'LOS', 'NT', 'CC']
```

```
internal_avg = [
    df['AWD_Int_Total'].mean(),
    df['WFS_Int_Total'].mean(),
    df['.NET_Int_Total'].mean(),
    df['LOS_Int_Total'].mean(),
    df['NT_Int_Total'].mean(),
    df['CC_Int_Total'].mean()
]
```

```
external_avg = [
    df['AWD_Ext_Total'].mean(),
    df['WFS_Ext_Total'].mean(),
    df['.NET_Ext_Total'].mean(),
    df['LOS_Ext_Total'].mean(),
    df['NT_Ext_Total'].mean(),
    df['CC_Ext_Total'].mean()
]
```

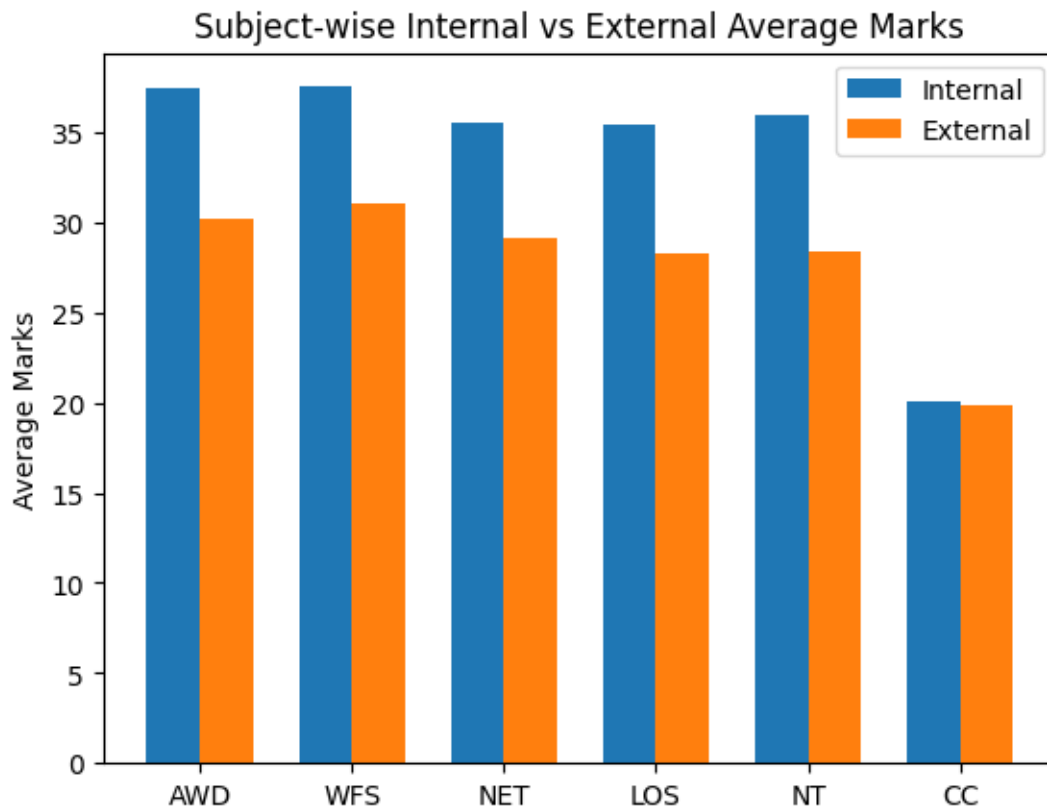
```
x = np.arange(len(subjects))
width = 0.35

plt.figure()

plt.bar(x-width/2, internal_avg, width, label='Internal Marks')
plt.bar(x+width/2, external_avg, width, label='External Marks')

plt.xticks(x, subjects)
plt.xlabel("Subjects")
plt.ylabel("Average Marks")
plt.title("Subject-wise Internal vs External Average Marks")

plt.legend()
plt.show()
```



Interpretation :

This visualization compares the average internal and external marks for each subject. The results help identify whether internal assessments are more lenient compared to university examinations. Significant differences between the two can indicate variations in evaluation patterns.

10.2 Correlation Heatmap of Subjects and SGPA :

```
import seaborn as sns
import matplotlib.pyplot as plt

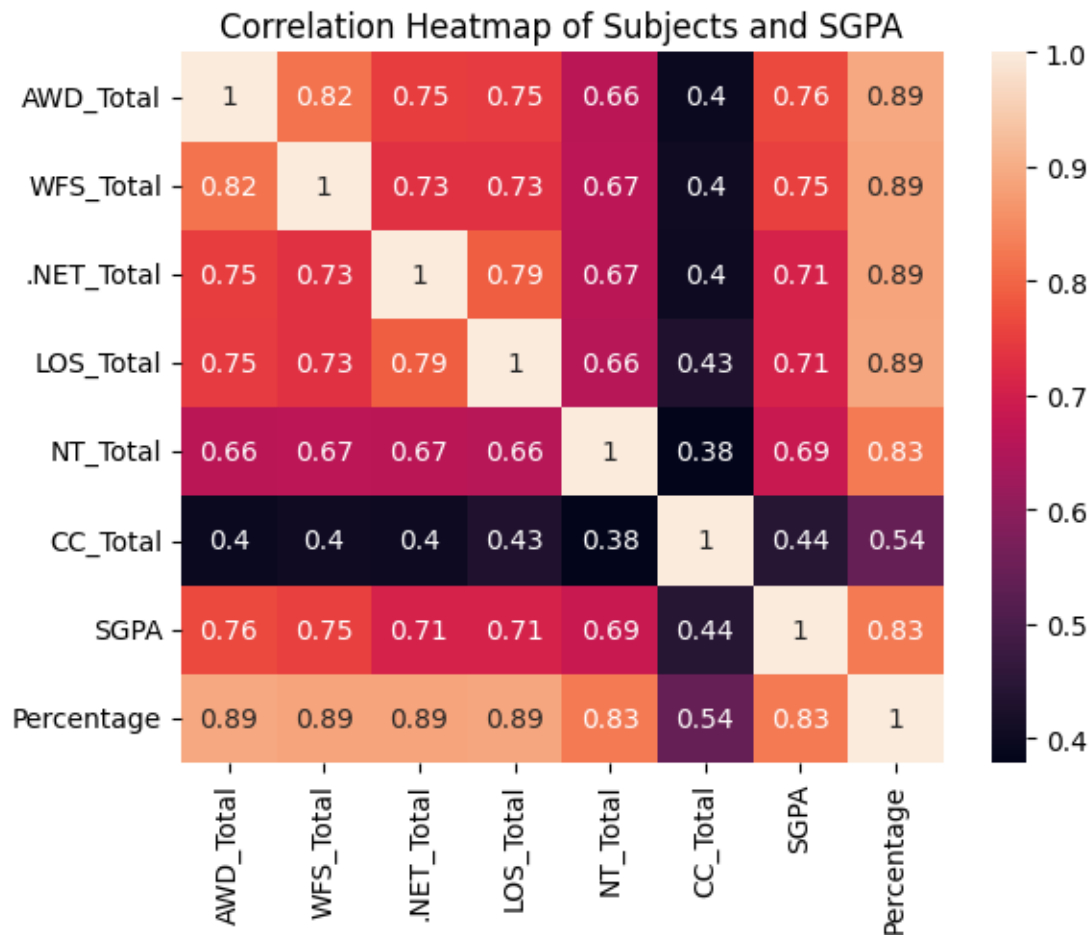
columns = [
    'AWD_Total', 'WFS_Total', '.NET_Total',
    'LOS_Total', 'NT_Total', 'CC_Total',
    'SGPA', 'Percentage'
]

plt.figure()

sns.heatmap(df[columns].corr(), annot=True)

plt.title("Correlation Heatmap of Subjects and SGPA")

plt.show()
```



Interpretation :

The heatmap illustrates the strength of relationships between subject marks, SGPA, and percentage. Higher positive correlations suggest that strong performance in those subjects contributes significantly to overall academic achievement.

10.3 Radar Chart of Subject Performance by Result :

```
import numpy as np
import matplotlib.pyplot as plt

subjects = ['AWD_Total', 'WFS_Total', '.NET_Total', 'LOS_Total', 'NT_Total', 'CC_Total']

pass_avg = df[df['Result']=='PASS'][subjects].mean().values
fail_avg = df[df['Result']!='PASS'][subjects].mean().values

angles = np.linspace(0, 2*np.pi, len(subjects), endpoint=False)

pass_avg = np.concatenate((pass_avg, [pass_avg[0]]))
fail_avg = np.concatenate((fail_avg, [fail_avg[0]]))
angles = np.concatenate((angles, [angles[0]]))

fig = plt.figure()
ax = fig.add_subplot(111, polar=True)

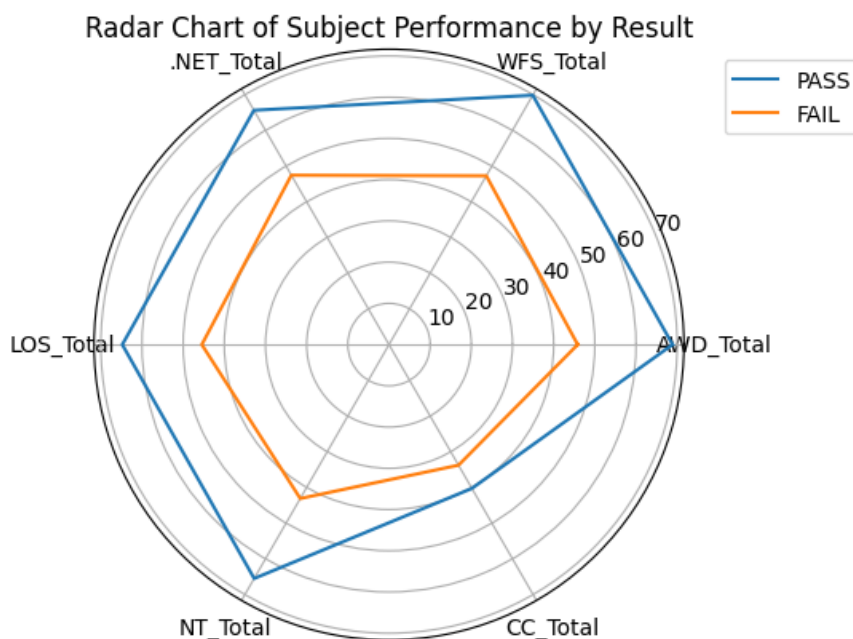
ax.plot(angles, pass_avg, label='PASS')
ax.plot(angles, fail_avg, label='FAIL')

ax.set_xticks(angles[:-1])
ax.set_xticklabels(subjects)

plt.title("Radar Chart of Subject Performance by Result")

# Move legend outside
plt.legend(loc='upper left', bbox_to_anchor=(1.05, 1))

plt.show()
```



Interpretation :

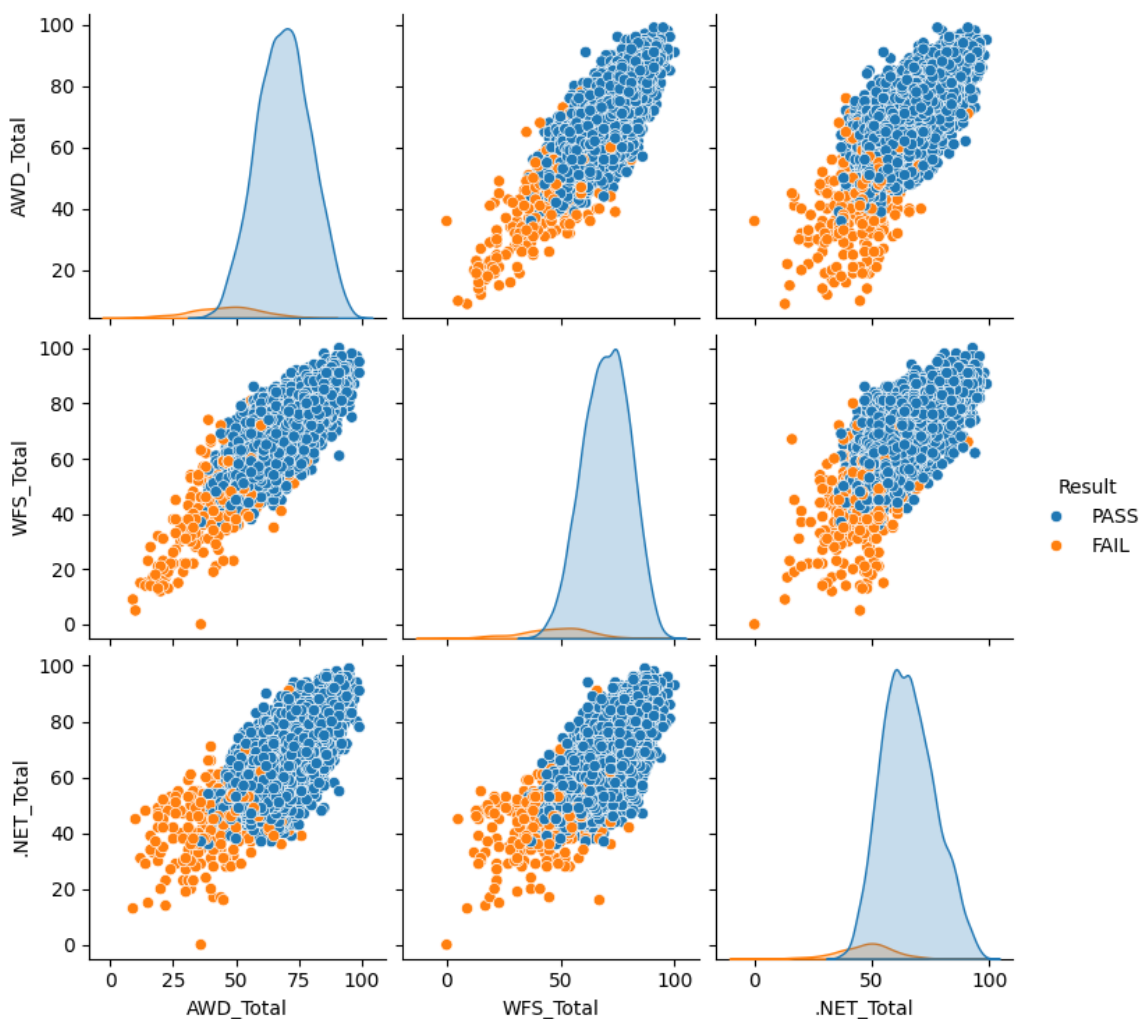
The radar chart visualizes differences in subject-wise performance between passing and failing students. Subjects where failing students score significantly lower indicate areas that may contribute most to academic failure.

10.4 Pair Plot (Scatter Matrix) of Core Subjects :

```
import seaborn as sns

sns.pairplot(
    df[['AWD_Total', 'WFS_Total', '.NET_Total', 'Result']],
    hue='Result'
)

plt.show()
```



Interpretation

The pair plot provides a comprehensive view of relationships between core subjects. The colour differentiation highlights clusters of passing and failing students, helping identify subject performance patterns associated with academic success.

11. Data Visualization Using Sweetviz

Data visualization plays a crucial role in understanding patterns, trends, and relationships within a dataset. Instead of manually creating multiple charts, the Sweetviz library was used to generate an automated exploratory data analysis (EDA) report. Sweetviz provides an interactive visualization dashboard that summarizes dataset characteristics including distributions, correlations, missing values, and variable relationships.

The generated report provides a comprehensive overview of student performance data in an easy-to-understand visual format.

11.1 Sweetviz Library Overview :

Sweetviz is a Python library designed for automated exploratory data analysis. It generates an interactive HTML report containing statistical summaries and visualizations for all variables in a dataset.

Key features include:

- Variable distribution visualization
- Target variable analysis
- Correlation analysis
- Missing value detection
- Feature comparison
- Automatic chart generation

This tool significantly reduces the effort required to manually create multiple charts for dataset exploration.

```
import sweetviz as sv

report = sv.analyze(df)

report.show_html("student_result_analysis.html")
```

Analysis of VNSGU Student Academic Performance using Python

Sweetviz
2.3.1

Get updates, docs & report issues here
Created & maintained by [Francois Bertrand](#)
Graphic design by [Jean-Francois Hains](#)

DataFrame

NO COMPARISON TARGET

8987 ROWS
10 DUPLICATES
6.1 MB RAM
42 FEATURES
3 CATEGORICAL
38 NUMERICAL
1 TEXT

ASSOCIATIONS

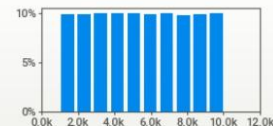
DataFrame

1 Seat Number

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 8,977 (>99%)
ZEROES: ---

MAX: 10,042
95%: 9,592
Q3: 7,772
AVG: 5,520
MEDIAN: 5,513
Q1: 3,264
5%: 1,455
MIN: 1,001

RANGE: 9,041
IQR: 4,507
STD: 2,609
VAR: 6.8M
KURT: -1.20
SKEW: 0.003
SUM: 49.6M



2 Name

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 8,965 (>99%)

2 <1% PATEL PRINCE NILESHBHAI
2 <1% JOGANI MITESH BHUPATBHAI
2 <1% RADADIYA PRINCE ASHOKBHAI
2 <1% KAKADIYA AASTHA BHAVESHBHAI
2 <1% KADIYA MANSIBEN KETANBHAI
2 <1% DHAMELIYA PRINCEKUMAR BHARATBHAI
2 <1% KACHHADIYA PARTH DINESHBHAI
8,973 >99% (Other)

3 College Name

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 36 (<1%)

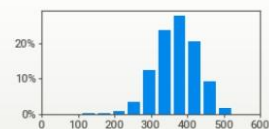
SDJ INTERNATIONAL COLLEGE, VESU, SURAT
MOTIRAM PATEL SCIENCE COLLEGE, BHARTHANA, VESU, SURAT
ILLEGE OF COMPUTER APPLICATION & SCIENCE, AMROLI, SURAT
IWALIBEN HARJIBHAI GONDALIYA COLLEGE OF BCA & IT, SURAT
(Other)

4 Total Marks

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 349 (4%)
ZEROES: ---

MAX: 524
95%: 461
Q3: 410
MEDIAN: 371
AVG: 370
Q1: 332
5%: 277
MIN: 109

RANGE: 415
IQR: 78.0
STD: 57.2
VAR: 3,271
KURT: 0.278
SKEW: -0.285
SUM: 3.3M

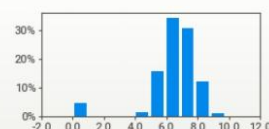


5 SGPA

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 64 (<1%)
ZEROES: 408 (5%)

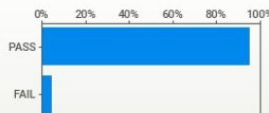
MAX: 9.82
95%: 8.36
Q3: 7.45
MEDIAN: 6.73
AVG: 6.47
Q1: 6.00
5%: 4.55
MIN: 0.00

RANGE: 9.82
IQR: 1.45
STD: 1.69
VAR: 2.86
KURT: 7.07
SKEW: -2.37
SUM: 58,133



6 Result

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 2 (<1%)



7 Course_Name

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 23 (<1%)

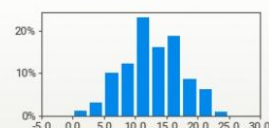
FUNDAMENTAL OF SOFTWARE ENGINEERING
CAREER READINESS AND PLACEMENT SKILLS
GIT FOR VERSION CONTROL
ADVANCED DATABASE SYSTEMS
(Other)

8 AWD_Ext_Th

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 26 (<1%)
ZEROES: 37 (<1%)

MAX: 25.0
95%: 20.0
Q3: 16.0
MEDIAN: 13.0
AVG: 12.6
Q1: 9.0
5%: 5.0
MIN: 0.0

RANGE: 25.0
IQR: 7.00
STD: 4.65
VAR: 21.6
KURT: -0.419
SKEW: -0.016
SUM: 113k



9 AWD_Ext_Pr

VALUES: 8,987 (100%)
MISSING: ---

MAX: 25.0
95%: 23.0

RANGE: 25.0
IQR: 4.00

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Analysis of VNSGU Student Academic Performance using Python



Analysis of VNSGU Student Academic Performance using Python



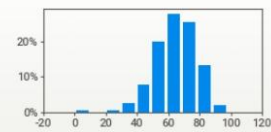
Analysis of VNSGU Student Academic Performance using Python

38

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 81 (<1%)
ZEROS: 49 (<1%)

MAX: 98.0
95%: 85.0
Q3: 74.0
MEDIAN: 65.0
AVG: 64.5
Q1: 56.0
5%: 42.0
MIN: 0.0

RANGE: 98.0
IQR: 18.0
STD: 13.8
VAR: 192
KURT: 1.78
SKEW: -0.730
SUM: 580k



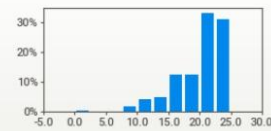
39

CC_Ext_Total

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 23 (<1%)
ZEROS: 31 (<1%)

MAX: 25.0
95%: 25.0
Q3: 23.0
MEDIAN: 21.0
AVG: 19.9
Q1: 18.0
5%: 12.0
MIN: 0.0

RANGE: 25.0
IQR: 5.00
STD: 4.11
VAR: 16.9
KURT: 1.75
SKEW: -1.21
SUM: 179k



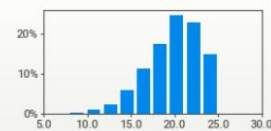
40

CC_Int_Total

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 19 (<1%)
ZEROS: ---

MAX: 25.0
95%: 25.0
Q3: 23.0
MEDIAN: 20.0
AVG: 20.1
Q1: 18.0
5%: 14.0
MIN: 6.0

RANGE: 19.0
IQR: 5.00
STD: 3.25
VAR: 10.6
KURT: 0.188
SKEW: -0.682
SUM: 180k



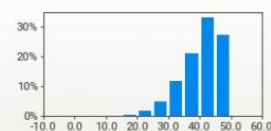
41

CC_Total

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 38 (<1%)
ZEROS: 1 (<1%)

MAX: 50.0
95%: 48.0
Q3: 45.0
MEDIAN: 41.0
AVG: 40.0
Q1: 36.0
5%: 28.0
MIN: 0.0

RANGE: 50.0
IQR: 9.00
STD: 6.37
VAR: 40.6
KURT: 0.663
SKEW: -0.890
SUM: 359k



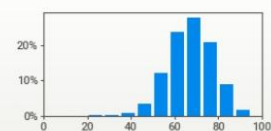
42

Percentage

VALUES: 8,987 (100%)
MISSING: ---
DISTINCT: 349 (4%)
ZEROS: ---

MAX: 95.3
95%: 83.8
Q3: 74.5
MEDIAN: 67.3
AVG: 67.3
Q1: 60.4
5%: 50.4
MIN: 19.8

RANGE: 75.5
IQR: 14.2
STD: 10.4
VAR: 108
KURT: 0.278
SKEW: -0.285
SUM: 605k



12. Machine Learning Implementation

In this chapter, the Python implementation used for student performance clustering is presented.

The code performs the following steps:

- Import required libraries
- Load cleaned dataset
- Check missing values
- Standardize features
- Apply K-Means Clustering
- Determine optimal clusters using Elbow Method
- Assign clusters to students
- Label clusters based on performance
- Visualize clusters using PCA
- Evaluate clustering using Silhouette Score
- Save trained model and results

12.1 Import Required Libraries :

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
import joblib
```

12.2 Load Dataset :

```
# Step 2: Load cleaned dataset

file_path = "D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\Data\\processed\\clea

df = pd.read_excel(file_path)

print(df.head())
print("Dataset Shape:", df.shape)
```


12.3 Select Features for Clustering :

```
feature_cols = [  
    "AWD_Total",  
    "WFS_Total",  
    ".NET_Total",  
    "LOS_Total",  
    "NT_Total",  
    "CC_Total"  
]  
  
X = df[feature_cols]
```

12.4 Check Missing Values :

```
print("Missing values in each column:")  
print(X.isnull().sum())
```

12.5 Standardize the Data :

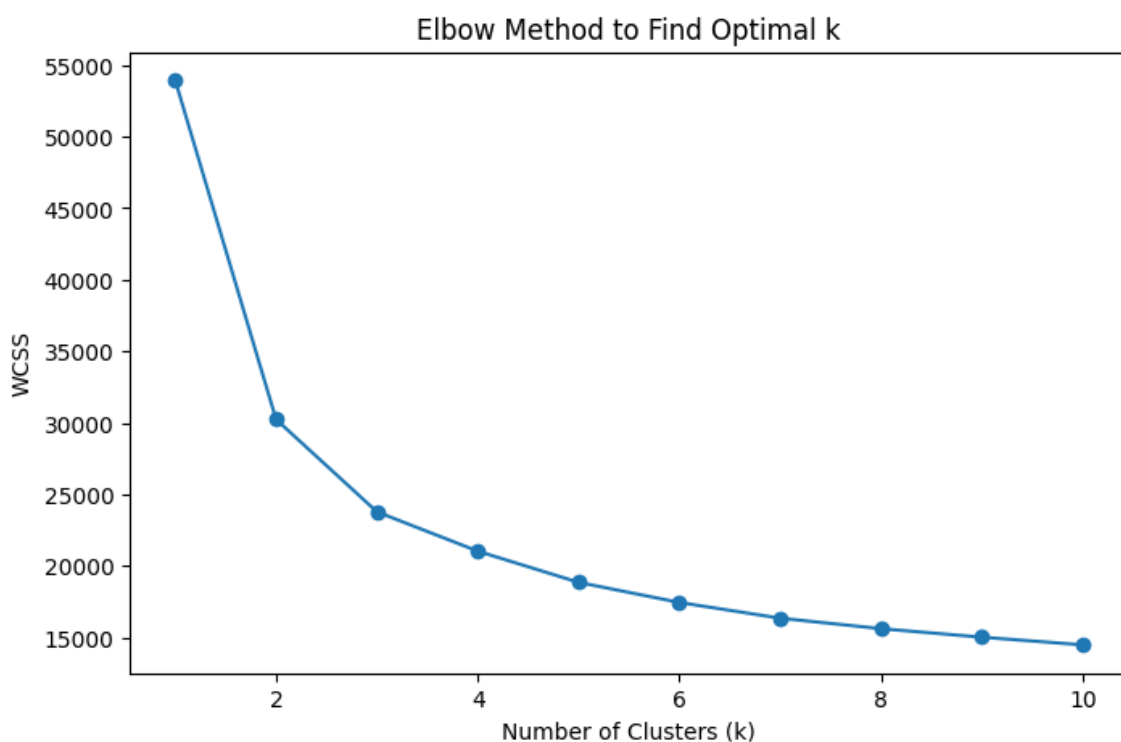
```
scaler = StandardScaler()  
  
X_scaled = scaler.fit_transform(X)  
  
X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)  
  
print("Mean of scaled features:")  
print(X_scaled_df.mean())  
  
print("Standard deviation of scaled features:")  
print(X_scaled_df.std())
```

12.6 Determine Optimal Clusters using Elbow Method :

```
wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method for Optimal Clusters")
plt.show()
```



12.7 Apply K-Means Clustering :

```
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

clusters = kmeans.fit_predict(X_scaled)

df["Cluster"] = clusters
```

12.8 Count Students in Each Cluster :

```
print(df["Cluster"].value_counts())
```

12.9 Cluster-wise Performance Summary :

```
cluster_summary = df.groupby("Cluster")[feature_cols].mean()

print(cluster_summary)
```

12.10 Assign Performance Labels :

```
performance_map = {
    0: "High Performer",
    1: "Average Performer",
    2: "Needs Improvement"
}

df["Performance_Level"] = df["Cluster"].map(performance_map)
```

12.11 PCA Visualization :

```
pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

pca_df = pd.DataFrame(X_pca, columns=["PC1", "PC2"])
pca_df["Cluster"] = df["Cluster"]
```

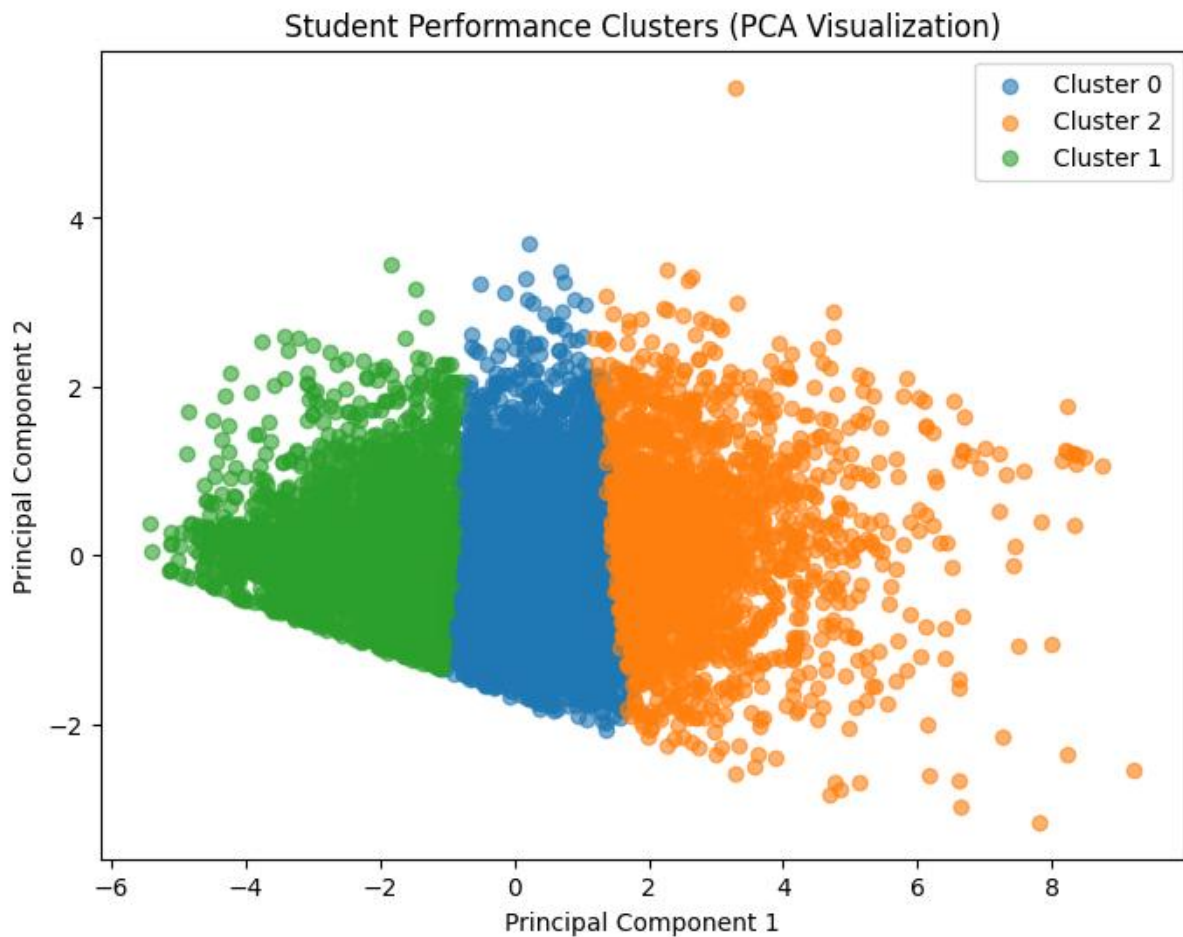
12.12 Plot Cluster Visualization :

```
plt.figure(figsize=(8,6))

for cluster in pca_df["Cluster"].unique():
    subset = pca_df[pca_df["Cluster"] == cluster]

    plt.scatter(
        subset["PC1"],
        subset["PC2"],
        label=f"Cluster {cluster}"
    )

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Student Performance Clusters (PCA Visualization)")
plt.legend()
plt.show()
```

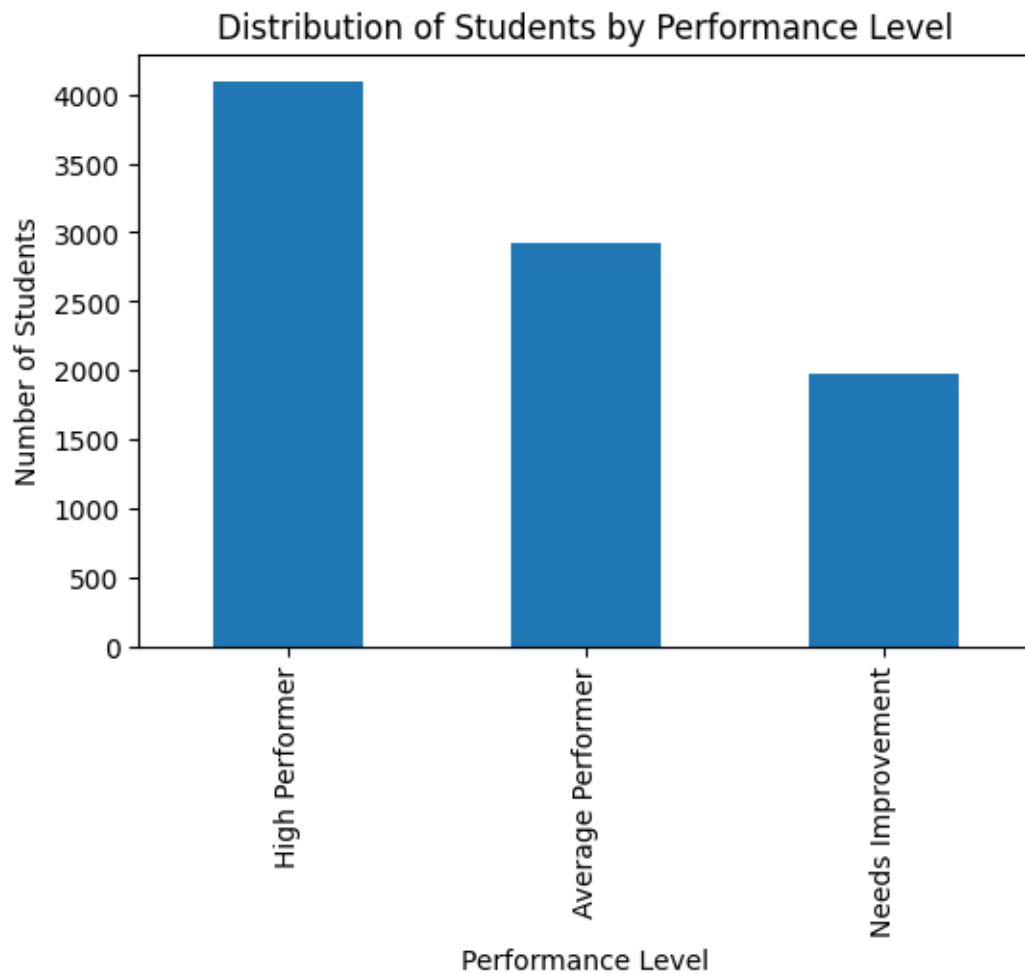


12.13 Performance Distribution Chart:

```
df["Performance_Level"].value_counts().plot(kind="bar", figsize=(6,4))

plt.xlabel("Performance Level")
plt.ylabel("Number of Students")
plt.title("Distribution of Students by Performance Level")

plt.show()
```



12.14 Evaluate Model using Silhouette Score :

```
score = silhouette_score(X_scaled, df["Cluster"])

print("Silhouette Score:", score)
```

12.15 Save Model and Results :

```
# Step 15: Save trained models

joblib.dump(
    kmeans,
    "D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\models\\kmeans_model.pkl"
)

joblib.dump(
    scaler,
    "D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\models\\scaler.pkl"
)

joblib.dump(
    pca,
    "D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\models\\pca_model.pkl"
)

print("Models saved successfully.")
```

12.16 Student Performance Prediction Script :

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
import joblib

# Load trained models
kmeans = joblib.load("D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\models\\kmea
scaler = joblib.load("D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\models\\scal
pca = joblib.load("D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\models\\pca_mod

# Take student marks as input
print("Enter Student Marks:")

new_student = {
    "AWD_Total": float(input("AWD Total: ")),
    "WFS_Total": float(input("WFS Total: ")),
    ".NET_Total": float(input(".NET Total: ")),
    "LOS_Total": float(input("LOS Total: ")),
    "NT_Total": float(input("NT Total: ")),
    "CC_Total": float(input("CC Total: "))
}
```

Analysis of VNSGU Student Academic Performance using Python

```
# Convert input into DataFrame
new_df = pd.DataFrame([new_student])

# Scale the input data
new_scaled = scaler.transform(new_df)

# Predict cluster
cluster = kmeans.predict(new_scaled)[0]

# Map cluster to performance label
performance_map = {
    1: "High Performer",
    0: "Average Performer",
    2: "Needs Improvement"
}

# Print prediction
print("\nPredicted Cluster:", cluster)
print("Performance Level:", performance_map[cluster])
```

```
PS D:\VNSGU_PDA_DATA\Data-Analytics-using-Python-Minor-Project\src> python predict.py
Enter Student Marks:
AWD Total: 89
WFS Total: 99
.NET Total: 92
LOS Total: 95
NT Total: 96
CC Total: 88
Performance Level: High Performer
```

```
PS D:\VNSGU_PDA_DATA\Data-Analytics-using-Python-Minor-Project\src> python predict.py
Enter Student Marks:
AWD Total: 65
WFS Total: 67
.NET Total: 68
LOS Total: 65
NT Total: 62
CC Total: 64
Performance Level: Average Performer
```

```
PS D:\VNSGU_PDA_DATA\Data-Analytics-using-Python-Minor-Project\src> python predict.py
Enter Student Marks:
AWD Total: 54
WFS Total: 56
.NET Total: 43
LOS Total: 45
NT Total: 56
CC Total: 21
Performance Level: Needs Improvement
```

13.Conclusion

By applying different data analysis techniques and based on the statistical and visual results, we have reached the following conclusions:

1. The majority of students fall within the average academic performance range, indicating balanced overall academic results.
2. Most students scored between 55% and 75%, which represents moderate to good academic performance.
3. The pass rate is significantly higher than the fail rate, showing satisfactory academic outcomes in the dataset.
4. Network Technology (NT) has the highest number of failures, indicating that it is comparatively the most difficult subject for students.
5. Cloud Computing (CC) has the lowest number of failures, suggesting stronger student performance in this subject.
6. A strong positive correlation exists between subject marks, SGPA, and percentage, indicating that good performance in individual subjects contributes significantly to overall academic success.
7. Students who passed generally obtained higher total marks and percentages, while failing students are concentrated in the lower score ranges.
8. The K-Means clustering model successfully categorized students into three performance groups: High Performer, Average Performer, and Needs Improvement, helping to identify academic performance patterns among students

14. References

- Python Official Documentation
<https://docs.python.org/3/>
- Pandas Official Documentation
<https://pandas.pydata.org/docs/>
- NumPy Official Documentation
<https://numpy.org/doc/>
- Seaborn Documentation
<https://seaborn.pydata.org/>
- Matplotlib Documentation
<https://matplotlib.org/stable/index.html>
- Scikit-learn Documentation
<https://scikit-learn.org/stable/>
- VNSGU Student Result Portal (Dataset Source)
<https://ums.vnsgu.net/Result/StudentResultDisplay.aspx?HtmlURL=7596,0000>
- UCI Machine Learning Repository
<https://archive.ics.uci.edu/>
- GeeksforGeeks – Machine Learning Tutorials
<https://www.geeksforgeeks.org/machine-learning/>